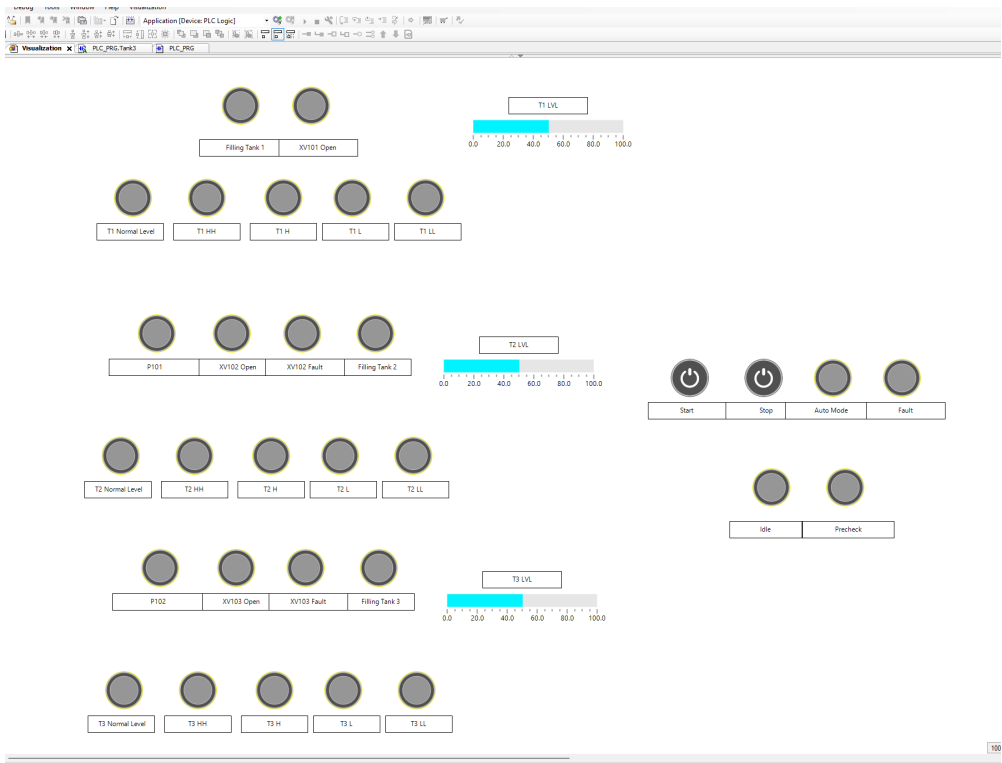


Three-Tank Water Transfer System

CODESYS PLC/HMI V1 Simulation Portfolio Report

Author: Mohammad Hamdan
Tools: CODESYS, Ladder Diagram, Structured Text, CODESYS Visualization
Status: Presentable V1 portfolio project



This report documents a CODESYS-based PLC/HMI simulation of a three-tank water-transfer process. The project demonstrates PLC sequence logic, pump and valve command-feedback control, simulated analog tank levels, alarm handling, manual operation, fault reset behavior, and HMI visualization.

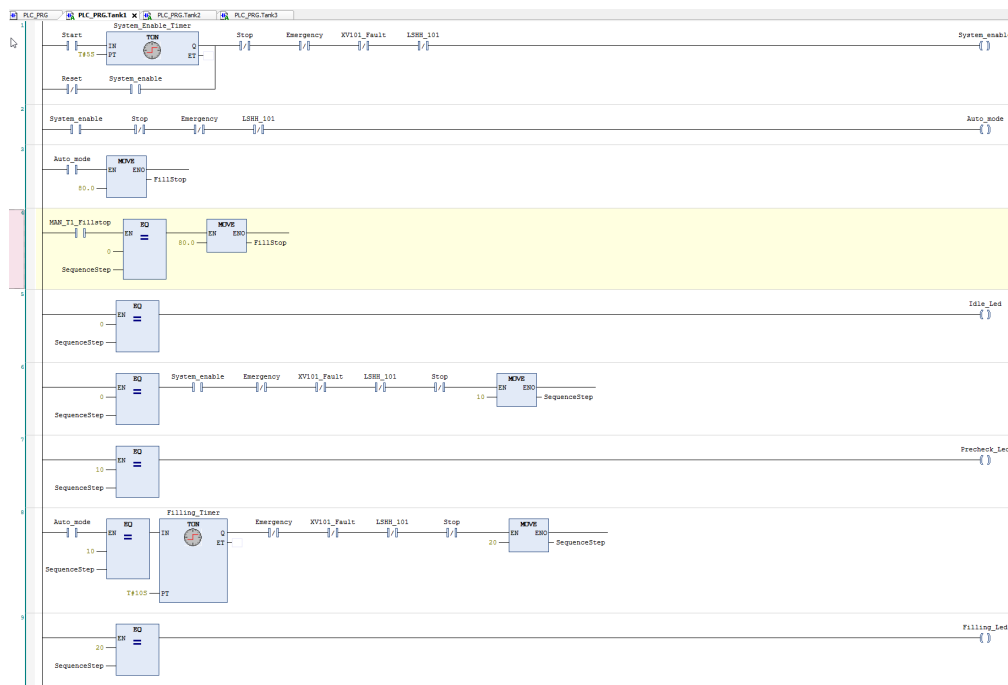
1. Project Overview

The V1 system fills T101 through an inlet valve, transfers water from T101 to T102 using valve XV102 and pump P101, and then transfers water from T102 to T103 using valve XV103 and pump P102. The process is implemented as an automatic step sequence with a fault state and reset handling.

Area	Implementation
PLC platform	CODESYS
Main logic	Ladder Diagram (LD)
Simulation logic	Structured Text (ST)
HMI	CODESYS Visualization
Project maturity	V1 simulation - portfolio ready after demo video

2. Automatic Sequence

Step	State	Description
S0	Idle	System waiting for start or reset state
S10	Precheck	Initial start and safety checks
S20	Fill T101	Open XV101 and fill Tank 101
S30	T101 Ready	Tank 101 reached fill setpoint
S40	Transfer T101 to T102	Open XV102 and run P101
S50	T102 Ready	Tank 102 reached fill setpoint
S60	Transfer T102 to T103	Open XV103 and run P102
S70	Complete	Process completed
S90	Fault	Fault state requiring reset



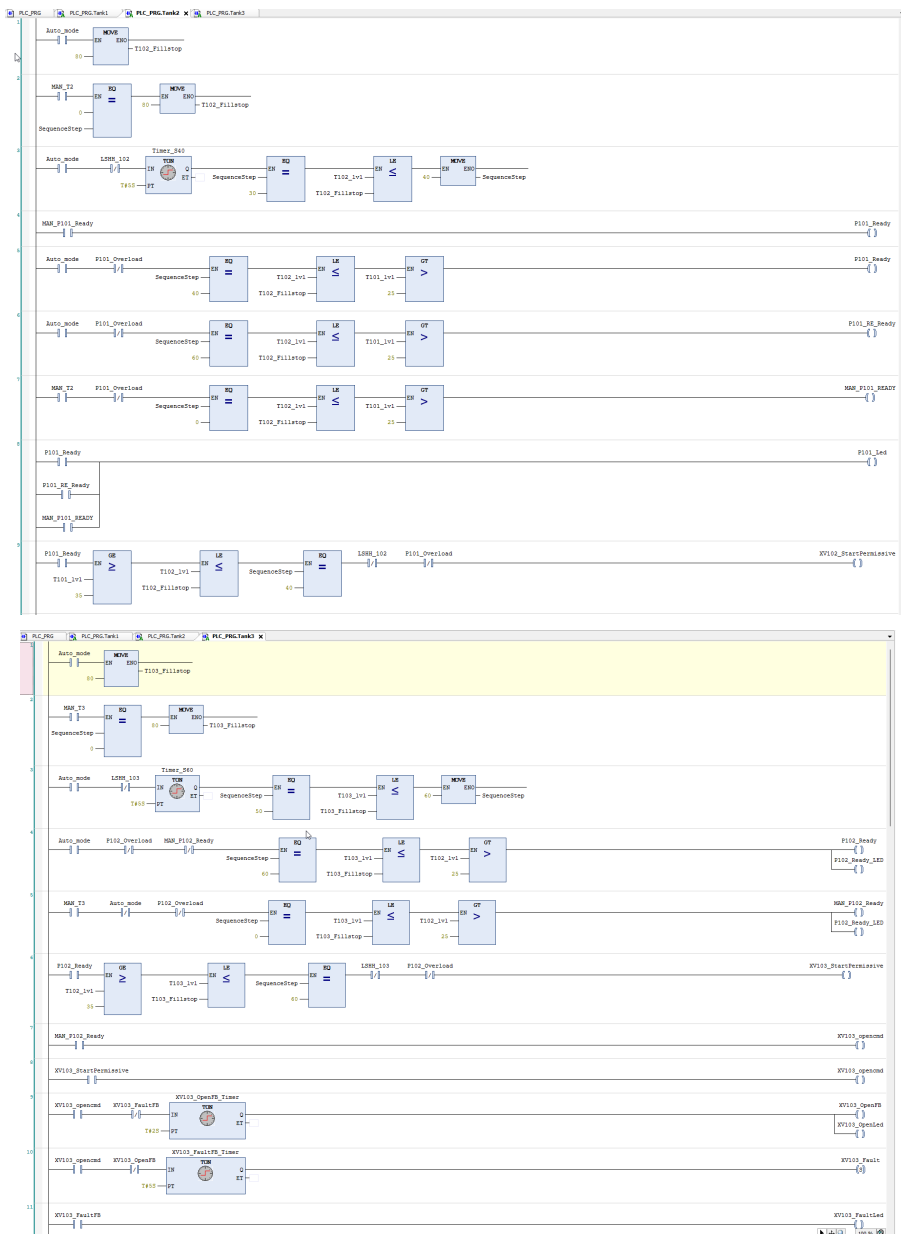
3. Device Command and Feedback Logic

The project follows the engineering pattern: permissive/request -> output command -> simulated feedback -> fault detection -> alarm indication. This is visible in the valve and pump logic for XV101, XV102, XV103, P101, and P102.



4. Transfer Logic

Transfer from T101 to T102 is controlled through XV102 and P101. Transfer from T102 to T103 is controlled through XV103 and P102. Each transfer checks source level, destination capacity, feedback, and fault/permissive conditions.



5. Structured Text Simulation

Structured Text simulates analog tank levels. T101 increases when XV101 feedback is active. During transfer, source tank levels decrease and destination tank levels increase. Each tank level is clamped between 0 and 100 percent.

PLC_PRG.Tank3	PLC_PRG	PLC_PRG.Tank2	PLC_PRG.Tank2_Sim_Level	PLC_PRG.Sim_Level x	PLC_PRG.Normal_Range
1	2	3	4	5	1
2	3	4	5	6	2
3	4	5	6	7	3
4	5	6	7	8	4
5	6	7	8	9	5
6	7	8	9	10	6
7	8	9	10	11	7
8	9	10	11	12	8
9	10	11	12	13	9
10	11	12	13	14	10
11	12	13	14	15	11
12	13	14	15	16	12
13	14	15	16	17	13
14	15	16	17	18	14
15	16	17	18	19	15
16	17	18	19	20	16
17	18	19	20	21	17
18	19	20	21	22	18
19	20	21	22	23	19
20	21	22	23	24	20
21	22	23	24	25	21
22	23	24	25	26	22
23	24	25	26	27	23
24	25	26	27	28	24
25	26	27	28	29	25
26	27	28	29	30	

```
//Fill T101 from inlet valve XV101
IF XV101_openFB AND (T101_lvl < FillStop) THEN
  T101_lvl := T101_lvl + 0.02;
END_IF;

// Transfer water from T101 to T102
IF P101_Run AND XV102_OpenFB AND (T102_lvl < T102_Fillstop) AND (T101_lvl > 35) THEN
  T101_lvl := T101_lvl - 0.01;
  T102_lvl := T102_lvl + 0.02;
END_IF

// Transfer water from T102 to T103
IF P102_Run AND XV103_OpenFB AND (T103_lvl < T103_Fillstop) AND (T102_lvl > 35) THEN
  T102_lvl := T102_lvl - 0.01;
  T103_lvl := T103_lvl + 0.02;
END_IF

//Limit T101
IF T101_lvl > 100.0 THEN T101_lvl := 100.0; END_IF;
IF T101_lvl < 0.0 THEN T101_lvl := 0.0; END_IF;

//Limit T102
IF T102_lvl > 100.0 THEN T102_lvl := 100.0; END_IF;
IF T102_lvl < 0.0 THEN T102_lvl := 0.0; END_IF;

//Limit T103
IF T103_lvl > 100.0 THEN T103_lvl := 100.0; END_IF;
IF T103_lvl < 0.0 THEN T103_lvl := 0.0; END_IF;
```

```
// T101 Range
IF (T101_lvl >= 25.0) AND (T101_lvl <= 80.0) THEN
  Normal_Level := TRUE;
ELSE
  Normal_Level := FALSE;
END_IF;

// T102 Range
IF (T102_lvl >= 25.0) AND (T102_lvl <= 80.0) THEN
  T102_Normal_Level := TRUE;
ELSE
  T102_Normal_Level := FALSE;
END_IF;

// T103 Range
IF (T103_lvl >= 25.0) AND (T103_lvl <= 80.0) THEN
  T103_Normal_Level := TRUE;
ELSE
  T103_Normal_Level := FALSE;
END_IF;
```

6. Future Improvements

Planned improvements for the next version include rebuilding/polishing the project in Siemens TIA Portal and WinCC, creating reusable function blocks for devices, improving manual mode structure, adding alarm history and trends, and completing a formal FAT report with screenshots.

7. Portfolio Positioning

This project should be presented as a CODESYS PLC/HMI V1 simulation project. It demonstrates practical PLC development ability and industrial automation thinking, while remaining honest that it is not yet a final commissioning package.